

Table 14: Full few-shot learning results on 5% training data. ‘-’ means that 5% time series is not sufficient to constitute a training set.

Methods	RDTU		Time-LLM		GPT4TS		DLinear		PatchTST		TimesNet		FEDformer		Autoformer		Stationary		ETSformer		LightTS		Informer		Reformer		
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
ETTh1	96	0.479	0.460	<u>0.483</u>	<u>0.464</u>	0.543	0.506	0.547	0.503	0.557	0.519	0.892	0.625	0.593	0.529	0.681	0.570	0.952	0.650	1.169	0.832	1.483	0.91	1.225	0.812	1.198	0.795
	192	0.624	0.536	<u>0.629</u>	<u>0.540</u>	0.748	0.580	0.720	0.604	0.711	0.570	0.940	0.665	0.652	0.563	0.725	0.602	0.943	0.645	1.221	0.853	1.525	0.93	1.249	0.828	1.273	0.853
	336	<u>0.741</u>	0.603	0.768	0.626	0.754	<u>0.595</u>	0.984	0.727	0.816	0.619	0.945	0.653	0.731	0.594	0.761	0.624	0.935	0.644	1.179	0.832	1.347	0.87	1.202	0.811	1.254	0.857
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	0.615	0.535	<u>0.627</u>	<u>0.543</u>	0.681	0.560	0.750	0.611	0.694	0.569	0.925	0.647	0.658	0.562	0.722	0.598	0.943	0.646	1.189	0.839	1.451	0.903	1.225	0.817	1.241	0.835
ETTh2	96	0.332	0.294	<u>0.336</u>	<u>0.397</u>	0.376	0.421	0.442	0.456	0.401	0.421	0.409	0.420	0.390	0.424	0.428	0.468	0.408	0.423	0.678	0.619	2.022	1.006	3.837	1.508	3.753	1.518
	192	0.405	0.423	<u>0.406</u>	<u>0.425</u>	0.418	0.441	0.617	0.542	0.452	0.455	0.483	0.464	0.457	0.465	0.496	0.504	0.497	0.468	0.645	0.697	3.534	1.348	3.975	1.933	3.516	1.473
	336	0.410	<u>0.438</u>	0.405	0.432	<u>0.408</u>	0.439	1.424	0.849	0.464	0.469	0.499	0.479	0.477	0.483	0.486	0.496	0.507	0.481	0.905	0.727	4.063	1.451	3.956	1.520	3.312	1.427
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	0.382	0.418	0.382	0.418	<u>0.400</u>	<u>0.433</u>	0.694	0.577	0.827	0.615	0.439	0.448	0.463	0.454	0.441	0.457	0.470	0.489	0.809	0.681	3.206	1.268	3.922	1.653	3.527	1.472
ETTm1	96	0.313	0.369	<u>0.316</u>	0.377	0.386	0.405	0.332	<u>0.374</u>	0.399	0.414	0.606	0.518	0.628	0.544	0.726	0.578	0.823	0.587	1.031	0.747	1.048	0.733	1.130	0.775	1.234	0.798
	192	<u>0.438</u>	<u>0.427</u>	0.450	0.464	0.440	0.438	0.358	0.390	0.441	0.436	0.681	0.539	0.666	0.566	0.750	0.591	0.844	0.591	1.087	0.766	1.097	0.756	1.150	0.788	1.287	0.839
	336	<u>0.446</u>	0.431	0.450	<u>0.424</u>	0.485	0.459	0.402	0.416	0.499	0.467	0.786	0.597	0.807	0.628	0.851	0.659	0.870	0.603	1.138	0.787	1.147	0.775	1.198	0.809	1.288	0.842
	720	<u>0.485</u>	<u>0.478</u>	0.483	0.471	0.577	0.499	0.511	0.489	0.767	0.587	0.796	0.593	0.822	0.633	0.857	0.655	0.893	0.611	1.245	0.831	1.200	0.799	1.175	0.794	1.247	0.828
	Avg	<u>0.421</u>	<u>0.426</u>	0.425	0.434	0.472	0.450	0.400	0.417	0.526	0.476	0.717	0.561	0.730	0.592	0.796	0.620	0.857	0.598	1.125	0.782	1.123	0.765	1.163	0.791	1.264	0.826
ETTm2	96	0.176	0.260	<u>0.174</u>	<u>0.261</u>	0.199	0.280	0.236	0.326	0.206	0.288	0.220	0.299	0.229	0.320	0.232	0.322	0.238	0.316	0.404	0.485	1.108	0.772	3.599	1.478	3.883	1.455
	192	0.213	0.286	<u>0.215</u>	<u>0.287</u>	0.256	0.316	0.306	0.373	0.264	0.324	0.311	0.361	0.394	0.361	0.291	0.357	0.298	0.349	0.479	0.521	1.317	0.850	3.578	1.475	3.553	1.484
	336	0.271	0.327	<u>0.273</u>	<u>0.330</u>	0.318	0.353	0.380	0.423	0.334	0.367	0.338	0.366	0.378	0.427	0.478	0.517	0.353	0.380	0.552	0.555	1.415	0.879	3.561	1.473	3.446	1.460
	720	<u>0.436</u>	<u>0.415</u>	0.433	0.412	0.460	0.436	0.674	0.543	0.454	0.432	0.509	0.465	0.523	0.510	0.553	0.538	0.475	0.445	0.701	0.627	1.822	0.984	3.896	1.533	3.445	1.460
	Avg	0.274	0.322	0.274	<u>0.323</u>	<u>0.308</u>	0.346	0.399	0.426	0.314	0.352	0.344	0.372	0.381	0.404	0.388	0.433	0.341	0.372	0.534	0.547	1.415	0.871	3.658	1.489	3.581	1.487
Weather	96	0.168	0.220	0.172	0.263	0.175	0.230	0.184	0.242	<u>0.171</u>	<u>0.224</u>	0.207	0.253	0.229	0.309	0.227	0.299	0.215	0.252	0.218	0.295	0.230	0.285	0.497	0.497	0.406	0.435
	192	0.220	0.267	<u>0.224</u>	<u>0.271</u>	0.227	0.276	0.228	0.283	0.228	0.277	0.272	0.307	0.265	0.317	0.278	0.333	0.290	0.307	0.294	0.331	0.274	0.323	0.620	0.457	0.446	0.450
	336	0.278	0.317	0.282	<u>0.321</u>	0.286	0.322	<u>0.279</u>	0.322	0.294	0.326	0.313	0.328	0.353	0.392	0.351	0.393	0.353	0.348	0.359	0.398	0.318	0.355	0.649	0.547	0.465	0.459
	720	0.367	0.384	<u>0.366</u>	<u>0.381</u>	<u>0.366</u>	0.379	0.364	0.388	0.384	0.387	0.400	0.385	0.391	0.394	0.387	0.389	0.452	0.407	0.461	0.461	0.401	0.418	0.570	0.522	0.471	0.468
	Avg	0.258	0.297	<u>0.260</u>	0.309	0.263	<u>0.301</u>	0.263	0.308	0.269	0.303	0.298	0.318	0.309	0.353	0.310	0.353	0.327	0.328	0.333	0.371	0.305	0.345	0.584	0.527	0.447	0.453
Electricity	96	0.143	0.238	0.147	<u>0.242</u>	0.143	0.241	0.150	0.251	<u>0.145</u>	0.244	0.315	0.389	0.235	0.322	0.297	0.367	0.424	0.518	0.697	0.638	0.639	0.609	1.265	0.919	1.414	0.855
	192	0.155	0.240	<u>0.158</u>	<u>0.241</u>	0.159	0.255	0.163	0.263	0.163	0.260	0.318	0.396	0.247	0.341	0.308	0.375	0.501	0.531	0.718	0.648	0.772	0.678	1.298	0.939	1.240	0.919
	336	0.174	0.273	0.178	0.277	0.179	<u>0.274</u>	<u>0.175</u>	0.278	0.183	0.281	0.340	0.415	0.267	0.356	0.354	0.411	0.574	0.578	0.758	0.667	0.901	0.745	1.302	0.942	1.253	0.921
	720	<u>0.220</u>	0.308	0.224	0.312	0.233	0.323	0.219	0.311	0.233	0.323	0.635	0.613	0.318	0.394	0.426	0.466	0.952	0.786	1.028	0.788	1.200	0.871	1.259	0.919	1.249	0.921
	Avg	0.173	0.265	0.179	<u>0.268</u>	0.178	0.273	<u>0.176</u>	0.275	0.181	0.277	0.402	0.453	0.266	0.353	0.346	0.404	0.627	0.603	0.800	0.685	0.878	0.725	1.281	0.929	1.289	0.904
Traf f ic	96	<u>0.410</u>	0.287	0.414	0.291	0.419	0.298	0.427	0.304	0.404	0.286	0.854	0.492	0.670	0.421	0.795	0.481	1.468	0.821	1.634	0.855	1.157	0.636	1.557	0.821	1.586	0.841
	192	<u>0.415</u>	0.290	0.419	0.291	0.434	0.305	0.447	0.315	0.412	<u>0.294</u>	0.894	0.517	0.653	0.405	0.737	0.503	1.509	0.838	1.856	0.928	1.688	0.848	1.596	0.834	1.602	0.844
	336	<u>0.439</u>	0.316	0.437	0.314	0.449	<u>0.313</u>	0.478	0.333	<u>0.439</u>	0.310	0.853	0.471	0.707	0.445	0.867	0.523	1.602	0.860	2.080	0.999	1.826	0.903	1.621	0.841	1.668	0.868
	720	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Avg	<u>0.421</u>	<u>0.298</u>	0.423	0.298	0.434	0.305	0.450	0.317	0.418	0.296	0.867	0.493	0.676	0.423	0.833	0.502	1.526	0.839	1.859	0.927	1.557	0.795	1.591	0.832	1.618	0.851
1 st Count	39		10		2		8		6		0		2		0		0		0		0		0		0		